

Performance Testing

Date	04 july 2026
Team ID	xxxxxx
Project Name	Human Development Index (HDI) Prediction Using Machine Learning
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman / Browser (Flask Local Server)
Type of Testing	Load Testing, Response Time Testing
Target Module / API	HDI Prediction Module (/predict)
Test Environment	Local Machine (Windows 11, Python 3.12, Flask)
Test Date	(02/07/2026)

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Open the Flask Home Page	1	30	Home page loads successfully
2	Predict HDI using valid input values	5	60	Correct HDI score and category displayed
4	Submit multiple prediction requests	10	120	Application responds without errors
	Continuous prediction requests	20	180	Stable performance with acceptable response time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 sec	0.82 sec	Pass	Fast response
2	Response Time (Maximum)	< 5 sec	1.76 sec	Pass	Within target
3	Throughput (Requests/sec)	> 5 req/sec	8 req/sec	Pass	Good performance
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	28%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	35%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

The Flask application loaded successfully without delays.

HDI predictions were generated accurately and quickly.

Multiple prediction requests were handled smoothly.

No crashes or performance degradation were observed during testing.

Bottlenecks Identified

No major bottlenecks were identified.

Performance may vary with very large datasets or heavy concurrent traffic.

Optimization Steps Taken

Optimized the Linear Regression model by using only the required input features.

Saved the trained model using model.pkl to avoid retraining during execution.

Used Flask for lightweight and efficient deployment.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]

Conclusion

The performance testing confirms that the Human Development Index (HDI) Prediction Using Machine Learning application performs efficiently under normal operating conditions. The application meets the expected response time, handles multiple prediction requests successfully, and maintains stable CPU and memory utilization. All performance test cases passed successfully.